CSA5108 Cryptography and Network security

Name : K. Sirisha
Reg.no:192325130

Exp 20 : Error Propagation in ECB and CBC Mode
Code:

```
#include <stdio.h>

int main(void) {
   int choice;

   printf("1. Error in transmitted ciphertext block C1\n");
   printf("2. Error in source plaintext block P1\n");
   printf("Enter choice: ");
   scanf("%d", &choice);

   if (choice == 1) {
      printf("\nCBC mode:\n");
      printf("P1 is corrupted because C1 is corrupted.\n");
      printf("P2 has corresponding bit errors because C1 is used in its
XOR.\n");
      printf("Blocks P3 and beyond are not affected by this
transmission error.\n");
   }
   else if (choice == 2) {
      printf("\nCBC mode:\n");
      printf("An error in P1 changes ciphertext C1.\n");
      printf("Because CBC is chained, subsequent ciphertext blocks
are also affected.\n");
      printf("The error can propagate through all remaining
ciphertext blocks.\n");
      printf("At the receiver, the affected plaintext blocks may be
corrupted.\n");
   }
   else {
      printf("Invalid choice.\n");
```

```
    }

    printf("\nECB comparison:\n");
    printf("A transmission error in one ciphertext block affects only
its corresponding plaintext block.\n");

    return 0;
}
```

